



Simultaneous NH/NO PLIF measurements in plasma-assisted ammonia and ammonia/hydrogen swirling flames

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ABSTRACT

This study investigates simultaneous NH/NO planar laser-induced fluorescence (PLIF) measurements in plasma-assisted NH₃/air and NH₃/H₂ (9:1 volume ratio)/air flames at equivalence ratios of $\phi = 0.75$, $\phi = 0.94$, and 1.1. A single dye laser system, equipped with frequency-doubling and mixing units, was employed to simultaneously generate excitation wavelengths near $\lambda_{\text{NO}} = 236.214 \text{ nm}$ and $\lambda_{\text{NH}} = 303.545 \text{ nm}$, enabling optimized excitation of NO and NH fluorescence, respectively. Across all equivalence ratios, plasma was found to enhance NO and NH concentrations in both NH₃/air and NH₃/H₂/air flames in the near field, although NH enhancement was less pronounced in the NH₃/H₂/air cases. In NH₃/air flames, NO concentrations decreased faster downstream with plasma activation, whereas in NH₃/H₂/air flames, NO levels remained relatively unchanged regardless of plasma activation. For NH₃/air flames, plasma could enhance atomic O production therefore acceleration of NH₃/NH₂/NH and form OH at the same time. The enhanced OH levels further promote NH production via $\text{NH}_2 + \text{OH} \rightarrow \text{NH} + \text{H}_2\text{O}$ in NH₃/air flames, though this effect is less significant in NH₃/H₂/air flames owing to the contribution of H₂ on radical pool buildup and less NH₃ availability in NH₃/H₂/air mixtures. In the downstream region, the reaction $\text{NH} + \text{NO} \rightarrow \text{N}_2\text{H}_2 + \text{H}$ plays a key role in reducing NO emissions in NH₃/air flames with plasma activation. These findings provide new insights into plasma-enhanced NH₃ flame chemistry and pollutant formation pathways, contributing to the development of cleaner and more efficient NH₃-based combustion technologies.

1. Introduction

Ammonia (NH₃) has emerged as a promising carbon-free fuel, particularly for marine transportation and energy production. Compared to hydrogen (H₂), NH₃ offers significant advantages in storage and transportation, as it can be easily liquefied and distributed. However, several challenges hinder its widespread adoption as a fuel, including its low laminar flame speed and the need to balance NH₃ and H₂ slip with NO_x and N₂O emissions. Addressing these challenges requires innovative approaches and a deeper understanding of NH₃ combustion mechanisms. Various strategies have been explored to improve NH₃ combustion, including fuel blending [1] and rich-quench-lean (RQL) staged combustion [2]. While these methods have shown potential, none have achieved both low emissions and enhanced flame stability simultaneously. Consequently, there is an urgent need to develop novel combustion concepts and integrate them with existing strategies to further optimize NH₃ combustors.

Plasma technology has demonstrated significant potential in NH₃ combustion by enhancing flame stability and reducing NO_x emissions [3–5]. Recent studies have highlighted the beneficial effects of

plasma on improving laminar flame speed [6], extending stability limits [7], and reducing ignition delays [3]. Non-equilibrium plasma-assisted combustion has gained widespread recognition as an effective approach for stabilizing flames in fossil fuel combustion [8–10]. Notably, plasma may increase NO_x formation in methane/air flames [8]; however, plasma could reduce NO_x emissions in NH₃/air flames [3–5, 11], although the effectiveness is dependent on the equivalence ratio and the specific characteristics of the plasma used [11]. It highlights the distinct chemical pathways influenced by plasma in different fuel systems.

Despite the demonstrated benefits of plasma on flame stabilization and de-NO_x (decrease NO_x) processes, the underlying mechanisms remain unclear due to the lack of effective diagnostic techniques. Most current studies [12–16] rely on NH₂* chemiluminescence and OH and O laser-induced fluorescence (LIF) techniques. These studies indicate that plasma enhances NH₂*, OH, and O formation in NH₃/air flames [12,15,17]. Based on the NH₂* results, Choe et al. [17] highlighted the role of NH₂ in the de-NO_x process, and hypothesized that NO is consumed through reactions with NH₂: $\text{NO} + \text{NH}_2 \rightarrow$

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$\text{NNH} + \text{OH}$ and $\text{NH}_2 + \text{NO} \rightarrow \text{N}_2 + \text{H}_2\text{O}$ in the thermal de- NO_x process. Recent studies [3,18] have also emphasized the role of NH in facilitating de- NO_x process, through pathways such as $\text{NH} + \text{NO} \rightarrow \text{N}_2\text{O} + \text{H}$. Therefore, investigating the plasma effects on fuel radical concentrations in plasma-assisted NH_3 flames is of great value.

Detecting NH in NH_3 combustion is essential for understanding both plasma-assisted and conventional combustion processes, as NH plays a critical role in both NO formation and consumption pathways [19]. Tang et al. [20] found that plasma increases NH formation and reduces the NH layer thickness in the near field of lean premixed NH_3/air flames. However, further investigation is needed to understand the interaction between NH and NO, requiring simultaneous detection of both species. Simultaneous NH and NO detection is also particularly valuable for unraveling NO_x emission mechanisms in NH_3 combustion. Additionally, studies on $\text{NH}_3/\text{H}_2/\text{air}$ flames and fuel rich flames remain scarce but results under these conditions could provide valuable information understanding the interaction between NH and NO and the role of plasma kinetics in their formation.

In this study, simultaneous NH/NO planar laser-induced fluorescence (PLIF) measurements were conducted in premixed NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames at equivalence ratios of 0.75, 0.94, and 1.1, both with and without plasma activation, using a swirl combustor. A single dye laser system was employed for PLIF excitation, with wavelengths tuned to $\lambda_{\text{NO}} = 236.214 \text{ nm}$ and $\lambda_{\text{NH}} = 303.545 \text{ nm}$. The influence of plasma on the spatial distributions of NH and NO was evaluated at various axial locations in both NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames. The kinetic effects induced by plasma activation in both flame configurations were subsequently analyzed and discussed.

2. Experimental methods

2.1. Burner setup and flame conditions

This work employed a swirl-stabilized dump combustor integrated with AC-powered discharges [20]. The combustor consisted of an annular axial swirler featuring 12 vanes with a trailing edge angle of 35° . The hub and outer swirler diameters were 11 mm and 19 mm respectively, resulting in a swirler number of 0.57 based on the definition provided by [8]. A square quartz tube measuring $70 \text{ mm} \times 70 \text{ mm}$ with a length of 305 mm was placed on top of the burner surface to confine the flames.

The combustor was equipped with a central electrode and an outer ring to generate plasma discharges using an AC power supply. The working voltage and current were monitored using an oscilloscope, showing peak values of 4000 V and 0.02 A, respectively, at a frequency of 20 kHz. The input energy was 30 W, corresponding to approximately 3% of the thermal power of the flame.

Fig. 1 presents the broadband chemiluminescence for flame cases at equivalence ratio of 0.75 (A1–A4), 0.94 (B1–B4), and 1.1 (C1–C4). For each equivalence ratios, four flame conditions are shown: NH_3/air and NH_3/H_2 (9:1 volume ratio)/air flames, each with and without plasma. A constant jet velocity of 3 m/s was maintained throughout the experiments. Without plasma, Flames A1 and B1 are slightly lifted, while Flame C1 shows more pronounced recirculation zones in the near field. When plasma is applied or H_2 is added, the flame attaches to the burner surface and the flame height is reduced. The white dashed and solid rectangles in Fig. 1 indicate the locations of the PLIF measurements. Red dots mark the positions of temperature measurements along the flame centerline using type-K thermocouples. These temperature data were mainly used to evaluate the uncertainty of the NO PLIF signal due to temperature variations in the downstream region.

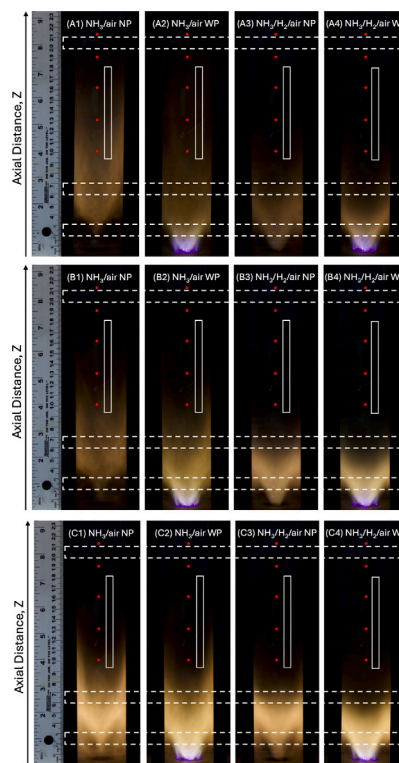


Fig. 1. Photographs of the flames captured using a DSLR camera (exposure time: 1/4 s, ISO 200) under the following conditions: (1) NH_3/air flame without plasma (NP), (2) NH_3/air flame with plasma (WP), (3) $\text{NH}_3/\text{H}_2/\text{air}$ flame without plasma, and (4) $\text{NH}_3/\text{H}_2/\text{air}$ flame with plasma. Each condition is shown at equivalence ratios of (A) $\phi = 0.75$, (B) $\phi = 0.94$, and (C) $\phi = 1.1$. The locations of the horizontal and vertical PLIF measurements are indicated by white dashed and solid rectangles, respectively. Red dots denote thermocouple measurement positions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.2. Simultaneous NO/NH PLIF techniques

Fig. 2 presents the NH and NO PLIF setup. A nanosecond Nd:YAG laser (Continuum, Powerlite DLS8000) pumped a tunable dye laser (Scanmate 2) at 532 nm, delivering 300 mJ/pulse at 10 Hz. The dye mixture (Rhodamine 610:640 = 3:1) produced 607 nm output, which was frequency-doubled to 303 nm (4 mJ/pulse).

The 303 nm beam was mixed with 1064 nm using a KDP crystal to generate 236 nm for NO PLIF (0.5 mJ/pulse). A Pellin–Broca prism separated the 236 nm and the residual 303 nm beam. Optimal excitation wavelengths were selected to target specific transitions from the ground electronic states: λ_{NH} was excited via the $\text{A}^3\Pi - \text{X}^3\Sigma^-$ (1,0) transition at 303.545 nm, and λ_{NO} via the $\text{A}^2\Sigma^+ - \text{X}^2\Pi$ (0,1) transition at 236.214 nm [21].

The 303 nm beam for NH PLIF was shaped using a set of cylindrical lenses with focal lengths of -75 mm (f_1), 400 mm (f_2), and 150 mm (f_3) to form a laser sheet measuring 15 mm (vertical) $\times$ 60 mm (horizontal). For NO PLIF, a single cylindrical lens with a focal length of 500 mm was used to generate a 5 mm $\times$ 60 mm laser sheet. The dimensions of the laser sheets were selected based on the expected signal strengths of NH and NO, ensuring an SNR > 8 in the flame regions. Note that the radial region near the quartz tube wall were excluded from the analysis due to strong window-related optical interference, resulting in a total image width of 40 mm for the NH and NO PLIF measurements. The NO/NH laser sheet thickness was approximately 400 μm (FWHM) (see Fig. S.1). NH fluorescence was captured using an f/100 mm UV lens and ICCD (Pimax4, 200 ns gate) with 332–342 nm bandpass filters. NO images used an f/100 mm UV lens with intensified CMOS camera (NAC GX-3,

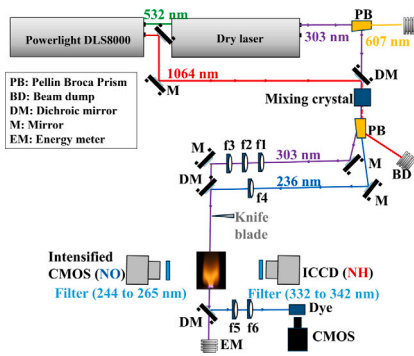


Fig. 2. Schematic of the experimental setup for simultaneous NH/NO PLIF measurements.

1 μ s gate) and 244–265 nm filters. The projected pixel size for NH and NO imaging was approximately 100 μ m/pixel.

During the experiments, the burner height was adjusted to access different axial locations (Z), enabling measurements at various downstream positions. At each location, 500 PLIF images were recorded. In addition, vertical NO/NH PLIF measurements were conducted by directing the laser from the burner top toward the burner surface to track the axial evolution of NO and NH from $Z = 12$ to 19 cm. Note that the laser pulses were not phase-locked with the AC plasma discharges given the quasi continuous nature of AC plasma discharge; thus, the PLIF measurements represent a random sampling across the discharge cycle. While species concentrations may vary within each cycle, the results remain valid for qualitative analysis of plasma effects. The root mean square (RMS) reported in this study reflects both spatial and temporal fluctuations.

2.3. Post processing

Images of NO/NH PLIF were post-processed as follows. The first step is background subtraction. The background is obtained by averaging 100 images taken with the laser blocked and the flame on. The images were then normalized by the pulsed energy recorded by the energy meter. Note the NH and NO PLIF signals were acquired within the linear excitation regime.

The spatial coordinates of the NO PLIF images were transformed to align with those of the NH PLIF images using a multi-dot target for image registration. Subsequently, the PLIF images were normalized by the vertical laser sheet profiles obtained from cuvette fluorescence images [20]. The NO laser sheet profile was corrected on a single-shot basis due to greater pulse-to-pulse variability, while the NH profile was corrected using an averaged cuvette image. The cuvette images were verified to exhibit a linear response with laser power. Spatial throughput non-uniformity was further corrected using camera images of a diffuser illuminated by a StellarNet uniform illumination sphere light source (UIS-LS-75 Watt). The estimated signal-to-noise ratio (SNR) for NH PLIF exceeded 12 in the flame reaction zones, while the SNR for NO PLIF was approximately 8 in the product regions, as verified in premixed $\text{NH}_3/\text{O}_2/\text{N}_2$ flames [20].

3. Results and discussion

Figs. 3 to 5 present simultaneous, instantaneous overlaid NO (blue) and NH (red) PLIF images at three axial segments: $Z = 20$ –35 mm, 60–75 mm, and 200–215 mm, shown in the first four columns. Note that the NO PLIF images cover a vertical height of 5 mm, while the NH PLIF images span 15 mm. The last two columns display the 1D ensemble-averaged PLIF intensity profiles extracted at $Z = 25$, 65, and 205 mm above the burner surface.

At $\phi = 0.75$, between $Z = 20$ and 35 mm, Flame A1 exhibits narrow distributions of NO and NH. The NO signal appears within the envelope of the NH layer, indicating that NO primarily forms on the product side of the flame, while NH marks the reaction zones. With plasma addition, NO formation increases and extends over a broader region. Simultaneously, NH intensity increases and displays more wrinkled structures, suggesting that plasma enhances combustion efficiency by expanding the NH-rich reaction zones. The addition of H_2 further increases NO concentrations and broadens the NO distribution. When both plasma and H_2 are applied, the NO region becomes even wider. Furthermore, with H_2 addition, NO is observed in both the inner and outer shear layers, likely due to enhanced recirculation zones.

The 1D profiles show the mean and root-mean-square (rms) values of NO and NH along the radial direction. For the NH_3 /air flame without plasma, both NO and NH are primarily confined to the central ± 10 mm region at $Z = 25$ mm. The central NO signal exhibits high rms values, indicating significant fluctuations in flame position. Plasma activation significantly increases the mean NO and NH intensity and broadens their spatial distribution. The peak NO intensity appears near ± 10 mm, while NH peaks slightly farther out, around ± 12 mm, with rms maxima occurring in similar regions. The broader spatial distribution and increased peak intensities of both NH and NO under plasma conditions suggest plasma could promote the formation of NH and NO concentrations near the flame base. For the NH_3/H_2 /air flame, the NO and NH peaks are located at ± 10 mm and ± 12 mm respectively—similar to the NH_3 /air flame with plasma. Plasma addition increases the NO peak intensity by approximately 1.5 times. While the NH peak intensity does not change significantly, the peak NH location shifts from ± 12 mm to ± 15 mm.

At $Z = 60$ to 75 mm, Flame A1 exhibits a broadband NO distribution enveloped by the NH layer. However, in cases with both plasma and H_2 addition (A2–A4), the NH signal becomes indistinct in this region due to the shortened flame structures. In NH_3 /air flame, the 1D NO profile shows that NO is primarily concentrated between -15 mm and $+15$ mm. With plasma addition, the NO distribution broadens. Although the rms remains low near the centerline, it increases toward the edges, indicating strong mixing between NO and combustion products in the outer regions. In the NH_3 /air flame, plasma reduces NO intensity, whereas in the NH_3/H_2 /air flame, plasma slightly enhances NO formation.

Due to the lifted flame structure without plasma activation, NO formation continues beyond $Z = 75$ mm. As a result, the NO concentration at $Z = 200$ –215 mm is comparable to, or slightly higher than, that at $Z = 60$ –75 mm near the centerline. At $Z = 200$ to 215 mm, the ensemble-averaged NO concentration in both NH_3/H_2 /air and NH_3 /air flames exhibits a parabolic distribution, with the centerline NO level approximately 1.5 times higher than that at the flame edges. In the NH_3 /air flame, plasma reduces NO intensity by a factor of 5, whereas in the NH_3/H_2 /air flame, the effect of plasma on NO emissions is minor, with changes of less than 15%.

Overall, plasma initially enhances both NO and NH concentrations in the near field, although the increase in NH_3/H_2 /air flames is less pronounced. Further downstream, the NO concentration in NH_3 /air flames decreases rapidly with the addition of plasma, reaching a significantly lower level at the exit ($Z = 205$ mm). In contrast, plasma has a minor effect on NO consumption in NH_3/H_2 /air flames, resulting in similar NO concentrations with and without plasma at the exit.

Fig. 4 presents the measurements for flames at $\phi = 0.94$ (B1–B4). Between $Z = 20$ and 35 mm, Flames B1–B4 exhibit behavior similar to A1–A4, where plasma activation enlarges both the NO and NH regions. This observation is further supported by the corresponding 1D ensemble-averaged profiles.

Moving downstream to $Z = 60$ –75 mm, NO exhibits a broadband distribution while the presence of NH segments interrupts the NO signal, resulting in a discontinuous structure. In contrast, for flames

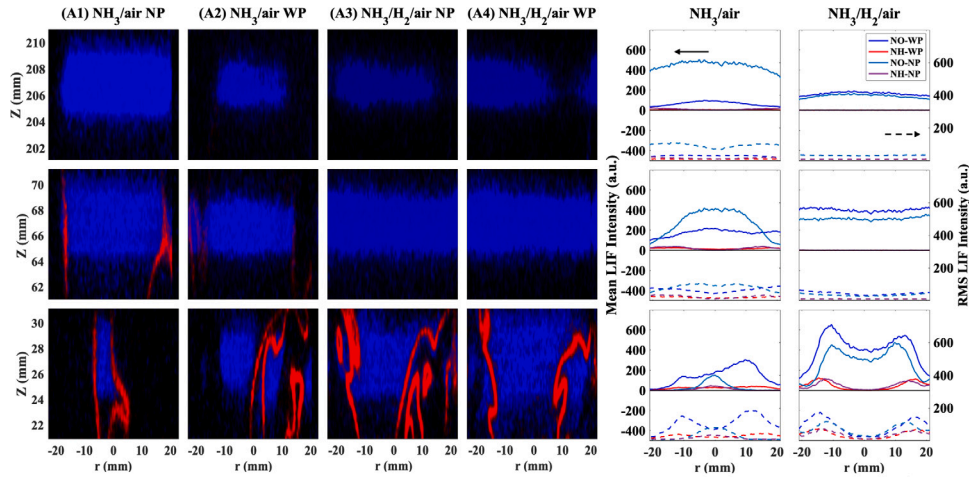


Fig. 3. Simultaneous single shot PLIF of NO (blue) and NH (red) for all four flame conditions (A1)–(A4) at $\phi = 0.75$. NO and NH 1D ensemble average (solid lines) and RMS (dashed lines) LIF intensities were calculated and plotted for NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames, with and without plasma. The 1D measurements were taken at $Z = 25$ mm, 65 mm, and 205 mm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

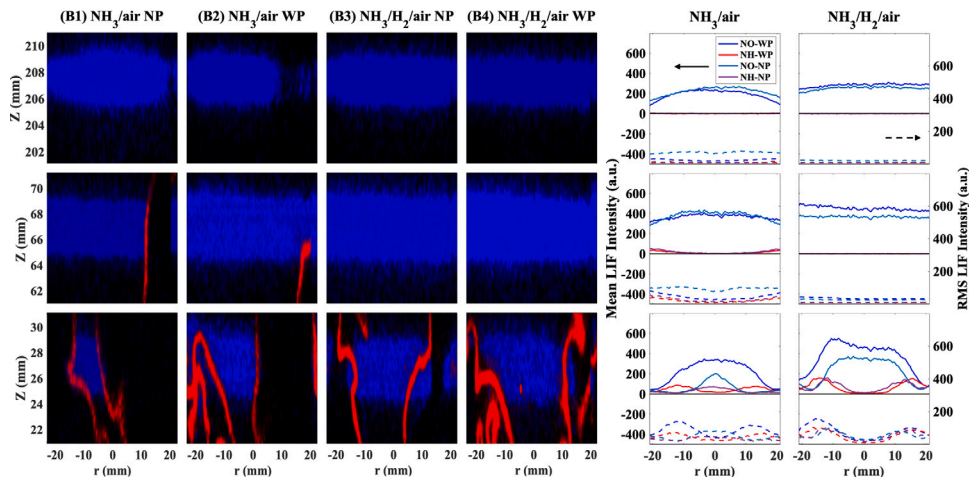


Fig. 4. Simultaneous single shot PLIF of NO (blue) and NH (red) for all four flame conditions (B1)–(B4) at $\phi = 0.94$. NO and NH 1D ensemble average (solid lines) and RMS (dashed lines) LIF intensities were calculated and plotted for NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames, with and without plasma. The 1D measurements were taken at $Z = 25$ mm, 65 mm, and 205 mm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

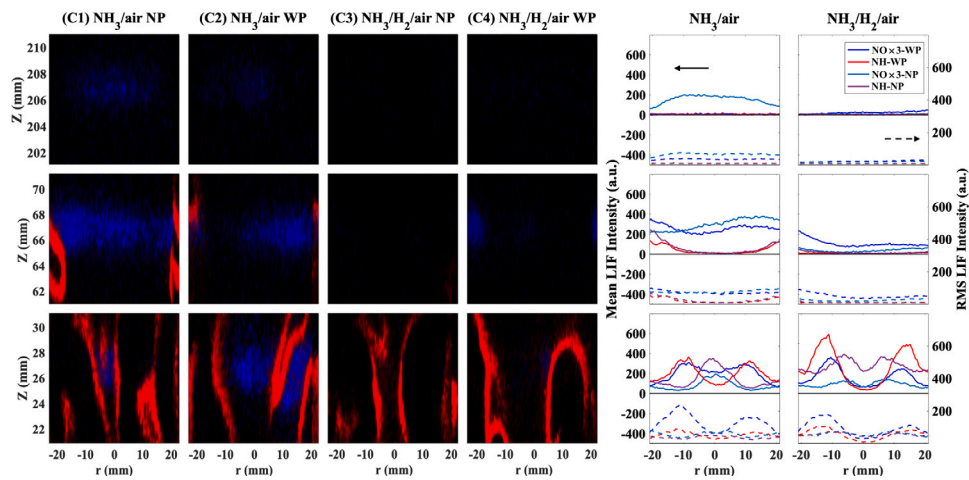


Fig. 5. Simultaneous single shot PLIF of NO (blue) and NH (red) for all four flame conditions (C1)–(C4) at $\phi = 1.1$. NO and NH 1D ensemble average (solid lines) and RMS (dashed lines) LIF intensities were calculated and plotted for NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames, with and without plasma. NO LIF was magnified by 3 times. The 1D measurements were taken at $Z = 25$ mm, 65 mm, and 205 mm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

B3 and B4, the NO signal remains nearly continuous, with no detectable NH. According to the 1D profiles, the NH_3/air flame exhibits a parabolic NO distribution, with similar NO structures observed in both the plasma and non-plasma cases. In the $\text{NH}_3/\text{H}_2/\text{air}$ flames, NO displays a more uniform distribution, and plasma addition clearly enhances the NO intensity.

At $Z = 200$ to 215 mm, the NO intensity decreases and becomes more discontinuous in the NH_3/air flames. However, in Flames B3 and B4, the NO signal remains spatially uniform. The ensemble-averaged NO intensity in the NH_3/air flame with plasma is lower, suggesting that plasma reduces NO formation near the flame exit. Interestingly, the NO intensity in the $\text{NH}_3/\text{H}_2/\text{air}$ flames is comparable to or even slightly higher than that of the flame with plasma, indicating that plasma has a distinct effect on NO formation when H_2 is present.

Fig. 5 presents the results for rich flames (C1–C4) in the near field where $Z = 20$ to 35 mm, where single shot NO signals are significantly lower than those in the lean cases, while NH intensity is notably stronger. In NH_3/air flames, plasma increases NO intensity in near field. In $\text{NH}_3/\text{H}_2/\text{air}$ flames, the single shot NO signal is close to the detection limit. The 1D radial profiles exhibit trends similar to those observed in the other flame series: both plasma activation and H_2 addition enhance NH and NO formation and shift their distributions toward larger radial positions. Further downstream, at $Z = 60$ to 75 mm, NO in NH_3/air flames decreases with plasma activation due to rapid NO consumption, while NH remains detectable near the wall. In $\text{NH}_3/\text{H}_2/\text{air}$ flames, NO levels are close to the noise threshold, but plasma leads to a slight increase in NO, again indicating a different trend compared to NH_3/air flames. The 1D profiles reveal a non-uniform NO distribution across all conditions.

At $Z = 200$ to 215 mm, NO in $\text{NH}_3/\text{H}_2/\text{air}$ flames remains near the noise level, regardless of plasma activation. Only the NH_3/air flame without plasma shows a detectable NO signal at the exit.

Overall, the simultaneous NO/NH PLIF measurements show consistent trends from lean to rich conditions. In NH_3/air flames, plasma enhances the formation of both NO and NH in the near field. However, due to the availability of NH , NO is consumed more rapidly downstream, resulting in lower NO emissions near the combustor exit. In $\text{NH}_3/\text{H}_2/\text{air}$ flames, plasma also promotes the formation of NO and NH, but the enhancement in NH is less significant. Further downstream, NO emissions remain similar with and without plasma. This is probably owing to the H_2 contributing to radical pool buildup, such as increased production of OH radicals, and therefore the effect of plasma is diminished. Another common observation from Figs. 3 to 5 is that NH and NO do not exist in the same area. It can be seen clearly from Figs. 3 to 5 that the NH layer separates the NO distribution regimes. It indicates that NH consumes NO quickly so they cannot co-exist in the same region.

Fig. 6 presents the ensemble-averaged NO profiles from $Z = 12$ to 19 cm, measured across a radial range from $r = 0.5$ to 1 cm (to avoid interference from center electrode and outer wall) and averaged accordingly. These measurements aim to track the axial evolution of NO in the downstream regions. NH is not shown, as its signal is negligible in these downstream regions. It is worth noting that the flame temperature variation for each equivalence ratio remains within 150 K from $Z = 12$ – 19 cm, based on thermocouple measurements. Details of these measurements are provided in the supplementary material. These temperature data indicate that the influence of the Boltzmann distribution on NO PLIF signal interpretation in this region is less than 5% [22]. Moreover, quenching effects from combustion products at the same equivalence ratio are not significant, as the variation in mole fraction remains within 3%. Therefore, neglecting the temperature sensitivity and quenching effects of NO LIF in the downstream region does not affect the conclusions drawn in this work.

In most conditions, the intensity of axial NO gradually decays beyond the main reaction zone (less than 20%); however, in the case of $\text{NH}_3/\text{H}_2/\text{air}$ WP at $\phi = 0.75$, a more significant drop (50%) is observed.

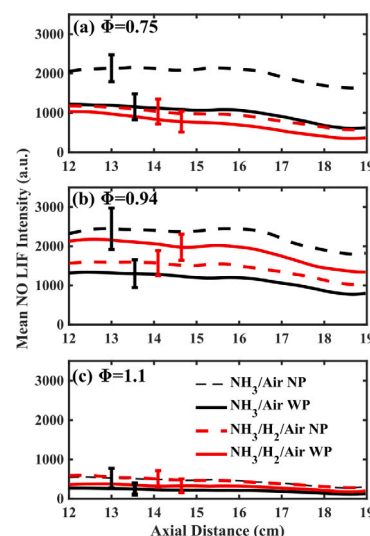


Fig. 6. Vertical distributions of NO obtained via laser-induced fluorescence (LIF) along the axial distance. Error bars represent one standard deviation.

NO emissions at $\phi = 0.75$ and $\phi = 0.94$ are very similar, whereas the flame at $\phi = 1.1$ shows significantly lower levels of NO.

In NH_3/air flames, plasma addition results in a substantial reduction of NO emissions in the downstream region, with NO intensity decreasing by more than 50%. In contrast, in NH_3/H_2 flames, plasma addition has a minimal effect on NO emissions and even leads to a slight increase, indicating distinct plasma kinetic effects in NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames.

Fig. 7 presents a sensitivity analysis of reaction pathways for NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames, performed using Chemkin-Pro in freely propagating flame mode. The simulation incorporates multicomponent transport and Soret effects while neglecting heat losses, and is based on the Stagni mechanism [23]. The analysis highlights the dominant reactions responsible for NO formation and consumption near the reaction zones.

As shown in Fig. 7(a) and (b), the chain-branching reaction $\text{O}_2 + \text{H} \rightarrow \text{O} + \text{OH}$ plays a key role in driving NO formation in both flame types. Same behavior were observed at $\phi = 0.75$ and $\phi = 1.1$ (shown in Fig. S.2). This is because the enhanced production of radicals, such as O and OH, promotes the oxidation of NH_3 to form more NH and NH_2 , which are key species for NO formation in NH_3 flames during the oxidation pathways of NH and NH_2 .

The activation of plasma and addition of H_2 may play similar roles to enhance radical production. Plasma is well-known to be effective on breaking O_2 molecules to form atomic O [14]. Atomic O then reacts with fuel molecules to enhance formation of NH and NH_2 through $\text{O} + \text{NH}_3 \rightarrow \text{NH}_2 + \text{OH}$ and $\text{O} + \text{NH}_2 \rightarrow \text{NH} + \text{OH}$ reactions. OH radicals subsequently enhance the oxidation of NH_3 , NH_2 , and NH to form more NO in the near field. With H_2 addition, the formation of radicals can be further enhanced through $\text{H}_2 + \text{OH} \rightarrow \text{H}_2\text{O} + \text{H}$ and then $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$ which is an overall chain branching process. These radicals react with NH_3 and its fragments, promoting NO formation [1]. Moreover, the increase in NH concentration is primarily driven by the reaction $\text{NH}_2 + \text{OH} \rightarrow \text{NH} + \text{H}_2\text{O}$ [1,20].

Moving downstream in NH_3/air flames, the enhancement of chain-branching reactions diminishes, and the reaction $\text{NH} + \text{NO} \rightarrow \text{N}_2\text{H}_2 + \text{H}$ becomes dominant, initiating a DeNO_x pathway. The plasma-generated NH radicals rapidly consume NO, resulting in lower NO concentrations further downstream.

In contrast, for $\text{NH}_3/\text{H}_2/\text{air}$ flames, plasma-induced NH radical formation is less significant. There are two possible reasons. First, less NH_3 is available at the same equivalence ratio condition with H_2 addition therefore, less NH is available comparing to the case without

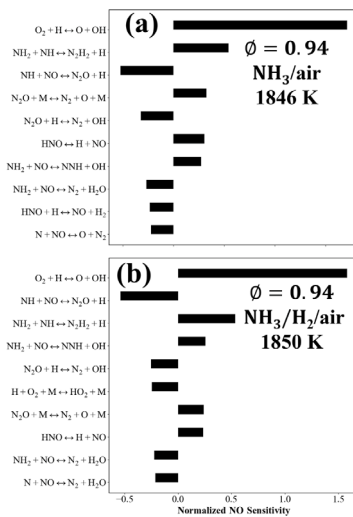


Fig. 7. Sensitivity analysis of NO production in 1D premixed flames at $\phi = 0.94$: (a) NH_3/air flame and (b) NH_3/H_2 (9:1 volume ratio)/air flame.

H_2 addition. Second, H_2 may consume a significant portion of atomic O radicals produced by the plasma diminishing the effect of plasma on NH_3 oxidation. The reaction between H_2 and atomic O, $\text{H}_2 + \text{O} \rightarrow \text{OH} + \text{H}$, produces H to enhance $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$ reaction. So the overall effect is to exchange one atomic O radical to three radicals (2 OH and one atomic O). Such an effect will significantly accelerate the oxidation of NH radicals and therefore less NH is available to consume NO downstream.

4. Conclusion

This study investigated the effects of plasma on NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames at $\phi = 0.75$, $\phi = 0.94$, and $\phi = 1.1$ through simultaneous NO and NH PLIF measurements. The results show that plasma enhances both NO and NH production in the near field for both NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames. However, the enhancement of NH is less pronounced in $\text{NH}_3/\text{H}_2/\text{air}$ flames. Further downstream, NO intensity decreases rapidly in NH_3/air flames with plasma, whereas NO levels remain similar in $\text{NH}_3/\text{H}_2/\text{air}$ flames with and without plasma.

For NH_3/air flames, radicals produced by plasma, such as atomic O, enhanced the oxidation of NH_3 which promoted the production of NO in the near field. The subsequently formed NH could further consume NO faster downstream through reaction $\text{NH} + \text{NO} \rightarrow \text{N}_2\text{H}_2 + \text{H}$. Therefore, NO concentration is higher in the near field but lower in the downstream. The overall effect is NO reduction by plasma.

For $\text{NH}_3/\text{H}_2/\text{air}$ flames, radicals produced by plasma, such as atomic O, might be consumed by H_2 resulting more OH production therefore enhance the oxidation of NH_3 and more NO formation in the near field. This effect is similar to the case of NH_3/air flame. However, the existence of H_2 reduced the availability of NH_3 therefore NH subsequently. This might be the reason why NO concentration remained constant downstream regardless of plasma activation. It was also observed that the existence of H_2 reduced the effect of plasma on the production of NH. This might be owing to the high reactivity of H_2 on radical production. The addition of H_2 in NH_3 significantly enhanced the $\text{H} + \text{O}_2 \rightarrow \text{O} + \text{OH}$ reaction as more H is available. This chain branching reaction diminished the effect of plasma on $\text{NH}_3/\text{H}_2/\text{air}$ flames.

Future work will include spatially resolved measurements of atomic oxygen to better understand the reaction pathways and NO formation mechanisms in NH_3 and H_3/H_2 flames under plasma activation.

CRediT authorship contribution statement

Hao Tang: Designed and performed research, Conceptualization, Analyzed data, Writing – original draft. **Evangelos Chatziandreou:** Performed research, Analyzed data, Writing – review & editing. **Griffin Rahn:** Performed research, Analyzed data, Writing – review & editing. **Bo Peng:** Performed research, Analyzed data, Writing – review & editing. **Wenting Sun:** Supervision, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary materials

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2025.105789>.

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